

TB Shimmer

DETAILED ENGLISH USER MANUAL

A granular shimmer and spatial-tail processor that turns windowed pitch-shifted grains into a controllable early-diffused, late-FDN cloud.



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Host-visible endpoints documented: 15

1. Product purpose

A granular shimmer and spatial-tail processor that turns windowed pitch-shifted grains into a controllable early-diffused, late-FDN cloud.

Conventional delay and reverb can add depth, but they do not automatically create the rising, reverse-blooming, particle-like motion associated with granular shimmer. Building that result from separate pitch, delay, feedback, filtering, diffusion, and randomization plug-ins is slow and difficult to recall. TB Shimmer combines those stages in one deterministic effect for pads, keys, guitar, vocals, percussion, loops, and sound-design sources while preserving a direct dry path.

What result it creates

- Pitch-rising or falling granular clouds without changing clip duration
- Distinct particles that can bloom into a smooth evolving stereo wash
- Independent control of grain timing, size, density, reversal, and three kinds of random variation
- Diffuse early reflections and a long filtered late tail with repeatable host recall

Signal flow

Input plus bounded filtered granular feedback -> 8-second history -> density-clocked Hann grains with pitch/reverse/randomization -> tone filtering -> raw/early-diffused blend -> direct wet plus eight-line late FDN -> M/S width -> Mix -> output

2. Complete feature guide

Time

Sets how far back each grain reads from the circular history, from 60 to 1200 ms. Short values stay close to the present source; long values separate the cloud from the attack. Time is a look-back position inside the effect and does not add reported host latency.

Size

Sets each grain's window length from 40 to 500 ms. Short grains sound more particulate and rhythmic; long grains overlap into smoother tones. Size also contributes to the smoothed early-diffusion blend and late-tail delay scale.

Density

Sets the grain spawn rate from 2 to 24 grains per second. Low density leaves audible events and gaps; high density builds a continuous cloud. Density also increases the early-diffusion blend and weights the late FDN more strongly.

Scatter

Varies grain size and stereo pan, and drives deterministic smooth-random motion in the diffuse tail. It controls spatial and temporal dispersion without replacing the dedicated Level Var, Position Rand, or Pitch Spray controls.

Pitch

Sets the base grain transposition from -12 to +24 semitones. +12 creates the classic octave shimmer; negative values create descending clouds; 0 preserves the source pitch before Pitch Spray is applied.

Reverse

Sets the probability that an individual grain plays backward. Low values add occasional inhaling motion; high values turn attacks into reverse blooms while the direct dry path remains available through Mix.

Feedback

Returns the filtered granular signal to history and independently lengthens the late diffuse tail, with a target decay up to about 45 seconds. High values accumulate pitch and energy, so raise it gradually and monitor the tail after transport stops.

Tone

Sets the upper spectral limit of the granular wet path from 1.2 to 18 kHz and damps every late-tail loop. Lower settings darken accumulated feedback and shorten high-frequency decay; higher settings preserve sparkle.

Width

Sets M/S width for the combined granular and diffuse wet signal from mono at 0 percent to deliberate widening at 200 percent. Recheck phase correlation and mono compatibility above 100 percent.

Mix

Linearly blends the sample-exact dry path with the complete granular, early-diffused, and late-tail signal. 0 percent is dry; 100 percent is wet only. Use a fully wet setting on an aux return.

Level Var

Randomly attenuates each grain by up to 12 dB. It never adds gain above the base grain level; higher values create a less uniform particle field and more depth.

Position Rand

Randomizes each grain's read position around the Time setting from 0 to 100 percent. Use it to loosen repeated timing without changing the overall delay center.

Pitch Spray

Adds independent bipolar pitch deviation around the Pitch setting by up to 12 semitones per grain. Small values add chorused instability; large values create atonal or cluster-like clouds.

Bypass

Uses a click-reduced host-visible bypass transition. Bypass removes the processed result for comparison; allow long tails to finish before removing the plug-in or closing the session.

Programs and state

The host-visible Program endpoint exposes 100 factory programs across utility, diffuse, sparkle, random, pulse, vocal, guitar, keys, pads, dark, and experimental groups. User presets and DAW state preserve all controls; legacy valid state is migrated to the current schema.

3. Quick start

1. Start from Init and set Mix to 100 percent temporarily so the wet structure is clear.
2. Set Pitch to +12 semitones for classic shimmer or choose another interval.
3. Balance Time, Size, and Density until the grain rhythm or smoothness fits the source.
4. Add Scatter, Reverse, Level Var, Position Rand, and Pitch Spray one at a time.

5. Raise Feedback only after the basic cloud is stable, then use Tone to control accumulated brightness.
6. Set Width and return Mix to the required insert or aux balance while checking mono compatibility and output headroom.

4. Practical workflows

Vocal halo

1. Set Pitch between +7 and +12 semitones.
2. Use medium Size and Density with modest Scatter.
3. Keep Reverse, Position Rand, and Pitch Spray low.
4. Darken Tone if sibilance builds and use moderate Feedback.
5. Blend at a conservative Mix or place fully wet on an aux.

Expected result: The vocal keeps intelligible attacks while a soft ascending halo blooms around phrase endings.

Pad nebula

1. Use long Time and Size with high Density.
2. Set Pitch to +12 or +19 semitones.
3. Increase Scatter, Position Rand, and Width.
4. Use medium Pitch Spray and Level Var for internal motion.
5. Raise Feedback for a long evolving tail and lower Tone if the cloud becomes brittle.

Expected result: Sustained harmony expands into a broad prismatic cloud that continues evolving between notes.

Reverse guitar bloom

1. Use medium Time and Size.
2. Raise Reverse strongly while keeping Density moderate.
3. Set Pitch to +12 and add a little Pitch Spray.
4. Use moderate Feedback and automate Mix into phrase endings.

Expected result: Picked attacks remain in the dry path while reversed octave particles swell behind them.

Percussive dust

1. Use short Size, short-to-medium Time, and low Density.
2. Keep Feedback low.
3. Increase Scatter, Level Var, and Position Rand.
4. Use a small Pitch Spray and reduce Mix.

Expected result: The groove gains scattered stereo particles without turning into a continuous reverb wash.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- High Feedback, bright Tone, upward Pitch, and dense grains can build energy quickly; lower Mix or Feedback and leave headroom.
- The late tail can remain audible for a long time after input stops; print or export beyond the last source event.
- Extreme Width may reduce mono compatibility, while dense long grains can soften transients and rhythmic definition.
- PULSE programs use free-running time values; the plug-in does not claim tempo synchronization or MIDI scale quantization.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.
- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 15 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
Bypass VST3 ID 773352680	Off, On	Off	Automatable; Bypass	Switches the host-visible click-reduced bypass transition.
Density VST3 ID 1552717032	2.0 to 24.0	9.0	Automatable	Sets grains spawned per second and also increases early-diffusion and late-tail weighting.
Feedback VST3 ID 1955982213	0.0 to 88.0	42.0	Automatable	Returns filtered granular wet signal to history and sets the independent diffuse-tail decay target.
Level Var VST3 ID 491597804	0.0 to 12.0	0.0	Automatable	Randomly attenuates individual grains by up to the selected dB amount without adding gain.
Mix VST3 ID 108124	0.0 to 100.0	48.0	Automatable	Linearly blends the sample-exact dry path with the complete granular and diffuse wet path.
Pitch VST3 ID 106677056	-12, -11, -10, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0, +1, +2, +3, +4, +5, +6, +7, +8, +9, +10, +11, +12, +13, +14, +15, +16, +17, +18, +19, +20, +21, +22, +23, +24	+12	Automatable	Sets the base transposition for every grain before per-grain Pitch Spray is applied.
Pitch Spray VST3 ID 1467082478	0.0 to 12.0	0.1	Automatable	Adds an independent bipolar random pitch offset around the base Pitch for each grain.
Position Rand VST3 ID 1066122619	0.0 to 100.0	32.0	Automatable	Randomizes each grain's history read position around the Time setting.

Parameter	Range / options	Default	Host status	Function
Program VST3 ID 1886548852	Init, Soft Halo, Open Air, Glass Halo, Guitar Ascension, Vocal Starlight, Pad Aurora, Slow Nebula, Wide Harmonic Fog, Reverse Comet, Falling Glass, Dark Orbit, Subtle Lift, Mono Bloom, Centered Glow, Bright Ambience, Warm Ambience, Transient Guard, Wet Shimmer Bus, Wet Diffuse Bus, Opal Chamber, Silken Room, Wide Diffusion, Warm Current, Dense Air, Slow Chamber, Short Bloom, Deep Diffusion, Silver Dust, Prism Ticks, High Constellation, Fifth Glint, Octave Sparks, Bright Rain, Fine Chimes, Star Shower, Gentle Variation, Loose Grains, Wandering Pitch, Soft Roulette, Broken Halo, Random Current, Wide Uncertainty, Total Scatter, Tight Glimmer, Broad Cloud, Staggered Drift, Slow Bloom, Quick Spark, Reverse Cadence, Slow Pulse, Long Cycle, Vocal Sheen, Vocal Octave, Vocal Air Trail, Vocal Velvet, Vocal Reverse Bloom, Vocal Choir Cloud, Vocal Shadow, Vocal Stars, Guitar Glow, Guitar Fifth, Guitar Octave Trail, Guitar Reverse Swell, Guitar Warm Cloud, Guitar High Mist, Guitar Low Orbit, Guitar Grain Storm, Piano Air, Piano Halo, Electric Silk, Organ Ascension, Synth Prism, Bell Mist, Keys Reverse Air, Low Key Cloud, Pad Soft Field, Pad Octave Wash, Pad Fifth Veil, Pad Reverse Tide, Pad High Horizon, Pad Low Horizon, Pad Random Choir, Pad Endless Light, Smoked Glass, Low Tide, Dim Chamber, Shadow Reverse, Black Velvet, Dusk Spray, Deep Random, Night Bloom, Frozen Fragments, Micro Swarm, Descending Mirage, Split Direction, No Pitch Cloud, Reverse Only Field, Shifting Constellation, Outer Limit	Init	Automatable	Selects one of 100 factory programs exposed to the host.
Reverse VST3 ID 1099846370	0.0 to 100.0	18.0	Automatable	Sets the probability that each grain reads its window backward.
Scatter VST3 ID 1911146174	0.0 to 100.0	32.0	Automatable	Varies grain size and stereo pan and drives deterministic smooth-random motion in the diffuse tail.
Size VST3 ID 3530753	40 to 500	160	Automatable	Sets grain duration and also influences the smoothed early-diffusion blend and late-tail delay scale.
Time VST3 ID 3560141	60 to 1200	240	Automatable	Sets the grain history look-back position; it is an internal creative delay and does not add reported host latency.
Tone VST3 ID 3565938	1200 to 18000	9000	Automatable	Low-passes the granular wet path and damps each pass through the late-tail feedback loop.
Width VST3 ID 113126854	0.0 to 200.0	135.0	Automatable	Sets M/S stereo width for the combined granular and diffuse wet signal.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.